A1: Voltage Controlled Oscillator (Ring vs. LC Tank) Analysis, Design, and Simulation

EN4431: Analog IC Design

March 2, 2026

Oscillators are autonomous circuits that generate a periodic output signal without any external AC input. A voltage-controlled oscillator (VCO) is a specialized type of oscillator where the output frequency is a linear function of an input control voltage.

In the context of a phase-locked loop (PLL), the VCO is the core engine. The PLL is a feedback system that compares the phase of the VCO output to a highly stable reference clock, adjusting the VCO's control voltage until the phases are locked. Because frequency is the derivative of phase, the VCO acts as an ideal integrator in the phase domain. Hence, the incremental phase domain relationship of a VCO is given by:

$$\Phi_{out}(t) = 2\pi K_{VCO} \int V_c(t) d(t) \tag{1}$$

Understanding both the physical circuit behavior and the Laplace-domain mathematical model of the VCO is critical for the stability analysis of the entire 2nd-order PLL system you will build during this course.

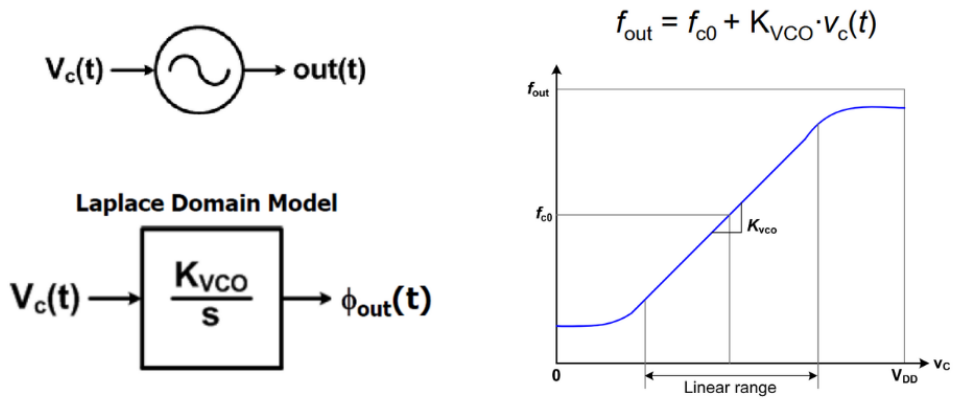


Figure 1: Abstract model of a voltage-controlled oscillator (VCO) [1]

1 Theoretical Analysis

Before designing the circuit in the SKY130 process, you must mathematically analyze the two dominant VCO topologies: the Ring oscillator and the LC tank oscillator.

1.1 Ring VCO Analysis

A ring oscillator consists of an odd number of inverting delay stages connected in a closed loop.

Q1.1.A State the Barkhausen criteria and prove how an odd number of identical delay stages satisfies the phase and gain requirements for oscillation.

Q1.1.B Derive the generic equation for the oscillation frequency (f_{osc}) in terms of the number of stages (N) and the delay per stage (t_d).

Q1.1.C For a current-starved inverter stage, derive the relationship between the control voltage (V_{ctrl}) and the delay (t_d).

Q1.1.D In a PLL, we track phase rather than frequency. Knowing that angular frequency ω is the time derivative of phase ϕ, derive the Laplace domain transfer function of the VCO:

$$H_{VCO}(s) = \frac{\Phi_{out}(s)}{V_{ctrl}(s)} \tag{2}$$

Express your final answer in terms of the VCO gain constant (K_{VCO}).

1.2 LC Tank VCO Analysis

Q1.2.A Derive the fundamental resonance frequency (ω_0) of an ideal parallel LC tank.

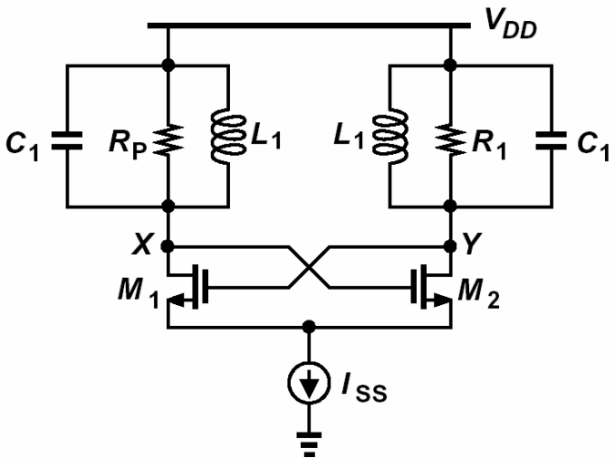


Figure 2: Cross-coupled LC VCO using NMOS pair

Q1.2.B Real inductors have parasitic series resistance, leading to energy loss. Explain the concept of "*negative resistance*" used to sustain oscillation. For a standard cross-coupled NMOS LC pair, derive the minimum transconductance (g_m) required to cancel a parallel tank resistance (R_p).

1.3 Task 1.3: Topology Comparison

Based on your analysis and literature review [1, 2], compile a comparison table summarizing the pros and cons of ring VCOs versus LC VCOs. Use the following parameters for comparisons:

- Silicon area
- Phase noise
- Tuning range
- Power consumption

2 Design and Simulation (Using Cadence tools & SKY130 PDK)

For designing, you will use a ring VCO topology due to its ease of integration and its wide tuning range.

2.1 Design a Current-Starved Ring VCO

Design a current-starved ring VCO at the schematic level using the SKY130 PDK. Begin with the simple current-starved inverter stage shown in Figure 3a. It is important to design the optimal inverter such that the charging and discharging phases are balanced, achieving a low settling time. Once the basic delay stage is complete, cascade multiple identical inverter stages to provide sufficient delay, and feed the final output back to the input. The complete current-starved ring VCO is shown in Figure 3b.

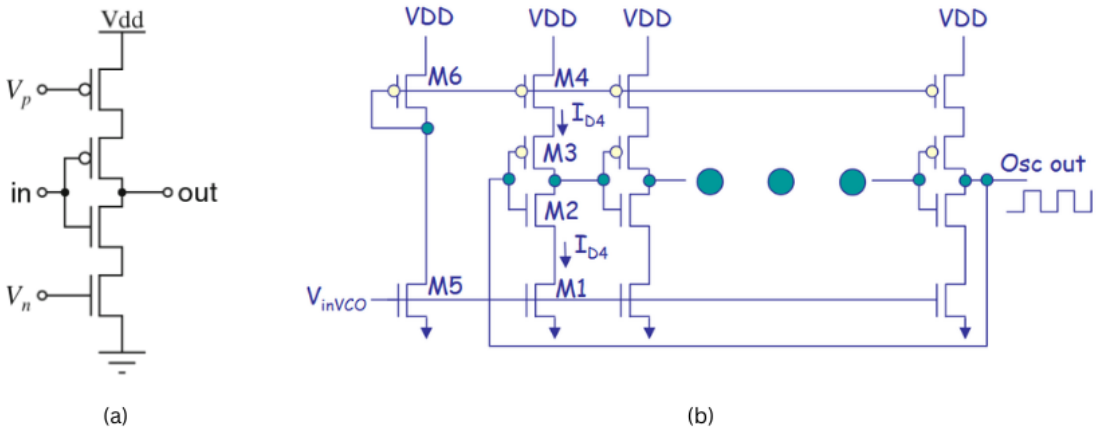


Figure 3: The circuit diagram of (a) a current-starved inverter stage and (b) a current-starved VCO

For more details on guidance on how to simulate and characterize the current-starved ring VCO, please refer to Design of CMOS Ring Oscillator - Chalmers University of Technology and ELEN 665 RF Communication Circuits – Laboratory 6: Analysis and Simulation of a CMOS VCO. The following required design specifications, summarized in the Table 1, are provided for this project,

Table 1: Desired Project Performance Specifications

Parameter	**Project Specification**
Technology [Min. length of transistors (L_{min})]	150 nm (SKY130A)
Supply voltage V_{DD}	1.8 V
Nominal input common-mode voltage $V_{DD}/2$	0.9 V
Load capacitance (C_{load})	50 fF
Oscillation center frequency (f_c)	100 MHz
Number of stages (N)	> 3 (justifiy the choice of N)
Tuning range	$> 50\%$ (i.e - 50 MHz to 150 MHz)
K_{VCO}	? MHz/V

2.2 Simulation and Characterization

To integrate this VCO into your final PLL, you must characterize its behavioral parameters to establish your 2nd-order system model. Perform the following simulations using Cadence Spectre:

Q2.2.A Run a basic transient simulation to verify startup and steady-state oscillation at the target center frequency (with V_{ctrl} set to a nominal midpoint voltage). Plot the output waveform.

Q2.2.B Perform a parametric sweep of V_{ctrl} from 0 V to 1.8 V. Plot the oscillation frequency (f_{osc}) vs. control voltage (V_{ctrl}).

Q2.2.C Using your tuning curve, calculate the VCO gain in the linear operating region:

$$K_{VCO} = \frac{\Delta f_{osc}}{\Delta V_{ctrl}} \quad \text{(Hz/V)} \tag{3}$$

Note: You must convert this to rad/s/V when plugging it into your Laplace domain model for the PLL loop filter design.

Q2.2.D Measure the average power dissipation of the core oscillator at the 100MHz center frequency.

Q2.2.E Characterize and report the measured performance metrics of your VCO design.

Q2.2.F Explicitly state the finalized linear transfer function of your simulated VCO ($H_{VCO}(s)$) that you will later pass forward to your PLL system-level model.

Instructions

Please submit your report (at least 5 pages without appendix), including the following information;

- Design methodology and details of the calculation and device sizes.
- Transient simulation results (up to 10 ns after steady state) and their frequency spectrum (FFT) showing the frequency of oscillation (f_{osc}) and sidebands.
- Parametric simulation results of frequency of oscillation (f_{osc}) vs. control voltage (V_{ctrl}).
- Phase noise (PS) and [Optional] Periodic steady state (PSS) simulation results.
- Discuss two improvements that can help achieve a wider, more stable tuning range for your design.
- Attach the circuit netlist in the appendix.

For more guidance, refer to the two videos and the ring VCO design document uploaded in the A1 drive folder.

Due Date: 15th March 2025 by 11.59 PM

Note - It is not essential for you to achieve the most optimal design with the best tuning range. If you can attempt to cover the most important simulations and relevant test benches for those simulations and present your work, it will be adequate to get the full assignment grades. Also, you have the freedom to improve the circuit as you wish with proper justifications.

Oscillator Phase Noise

Noise is injected into an oscillator by the devices that constitute the oscillator itself, including the active transistors and passive elements. This noise will disturb both the amplitude and frequency of oscillation. This is depicted in Figure 4, and more information is available in the LC tank voltage-controlled oscillator tutorial.

- Amplitude noise is usually unimportant because non-linearities that limit the amplitude of oscillation also stabilize the amplitude noise.
- Phase noise, on the other hand, is essentially a random deviation in frequency, which can also be viewed as a random variation in the zero crossing points of the waveform.

An oscillator is a frequency-selective (or narrow band, high-quality factor (Q)) circuit and will tend to reject out-of-band (i.e., frequencies offset from f_{osc} by Δf) signals to some degree. This rejection increases at larger offsets from f_{osc}. As a result, the effect of the noise sources is to produce "skirts" on either side of the ideal impulse function of the oscillator in the frequency domain. To measure or quantify phase noise, one considers a unit bandwidth at an offset Δf with respect to f_{osc} (i.e., relative offset frequency), and calculate the noise power in this bandwidth, and divide the result by the average carrier power.

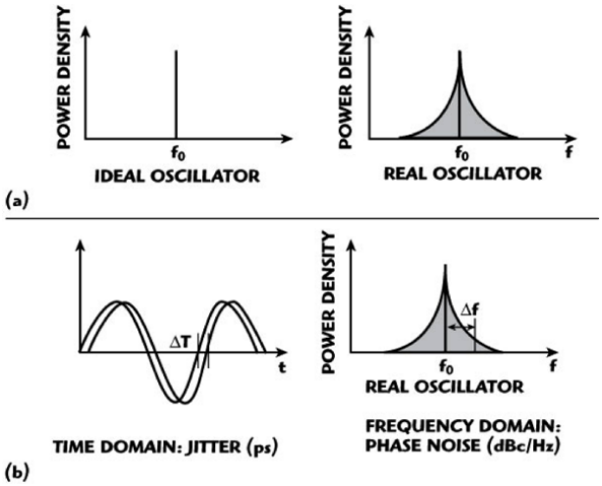


Figure 4: (a) Frequency spectrum of ideal and real oscillators. (b) Jitter in the time domain and phase noise in the frequency domain in a real oscillator

References

[1] S. Palermo, "Lecture 28: VCOs," ECEN689, Texas A&M University, 2010. [Online]. Available: https://people.engr.tamu.edu/spalermo/ecen689/lecture28_ee689_vcos.pdf

[2] M.-T. Hsieh and G. Sobelman, "Comparison of LC and Ring VCOs for PLLs in a 90 nm Digital CMOS Process," 01 2006.